



Automatic Water Dispenser

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A branded automatic water dispenser is usually expensive. So, a relatively low-cost do-it-yourself solution is presented here. It is a portable general-purpose automatic water dispenser perfectly suitable for homes to wash hands or feed water to pets. The presented system also offers many possibilities for extensions.

The system consists of three key segments: infrared (IR) proximity sensor, water pump driver, and mini submersible water pump. Also, there is a power supply block that holds a couple of standard dry cells.

IR proximity sensor design. The

use of an IR proximity sensor is a practical way to build an automatic water dispenser. While industrial-grade sensors are extremely efficient, most readymade pre-wired sensor modules are prone to iterated false triggering, which badly affects the intended performance. You also have to properly mount the sensors so that these are not exposed to strong sunlight/ambient light. These sensors should be covered partially to prevent erroneous operation.

To keep the complexity and cost within limits, a reliable IR proximity sensor has been developed with the help of a PLL tone decoder IC NE567.

This tone and frequency decoder is a highly stable phase-locked loop (PLL) with synchronous AM lock detection and power output circuitry. Its primary function is to drive a load whenever a sustained frequency within its detection band is present at the self-biased input. The bandwidth centre frequency and output delay are independently set

using just four external components.

The next key component is the compact TCRT5000L reflective sensor, which includes a 950nm IR emitter and phototransistor in a leaded package that blocks visible light.

Water pump driver. Any 3-6V water pump motor should work with this little driver built around transistor T2. (It is recommended to use a prime version water pump for a permanent fixture.) Just attach a small rubber/

Test Points: Measurement

Test points	Remarks
TP0	GND
TP1	1kHz-2kHz using VR1
TP2	Low when a valid signal is received at OS1
TP3	Low when there is an obstacle in front of TCRT5000, otherwise high
TP4	Around 6V

PARTS LIST

Semiconductors:

- | | |
|-----|---------------------------------------|
| IC1 | - NE567 tone decoder |
| OS1 | - TCRT5000L reflective optical sensor |

- | | |
|--------|--------------------------|
| T1 | - S8050 npn transistor |
| T2 | - S8550 pnp transistor |
| D1, D2 | - 1N4007 rectifier diode |
| LED1 | - 5mm LED |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|------------|------------------------|
| R1 | - 4.7-kilo-ohm |
| R2, R4, R5 | - 10-kilo-ohm |
| R3 | - 100-ohm |
| R6, R7 | - 1-kilo-ohm |
| VR1, VR2 | - 10-kilo-ohm potmeter |

Capacitors:

- | | |
|--------|---------------------------------|
| C1, C8 | - 100 μ F, 16V electrolytic |
| C2, C5 | - 100nF ceramic disk |
| C3 | - 2.2 μ F, 16V electrolytic |
| C4 | - 1 μ F, 16V electrolytic |
| C6 | - 10nF ceramic disk |
| C7 | - 220 μ F, 16V electrolytic |

Miscellaneous:

- CON1 - 2- pin connector
- M1 - 6V DC mini submersible pump motor
- 6V power supply/1.5V×4 AA or AAA cell

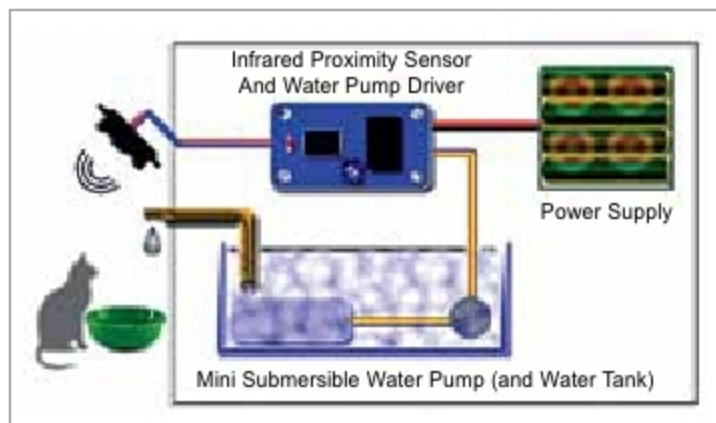


Fig. 1: Block diagram for system overview

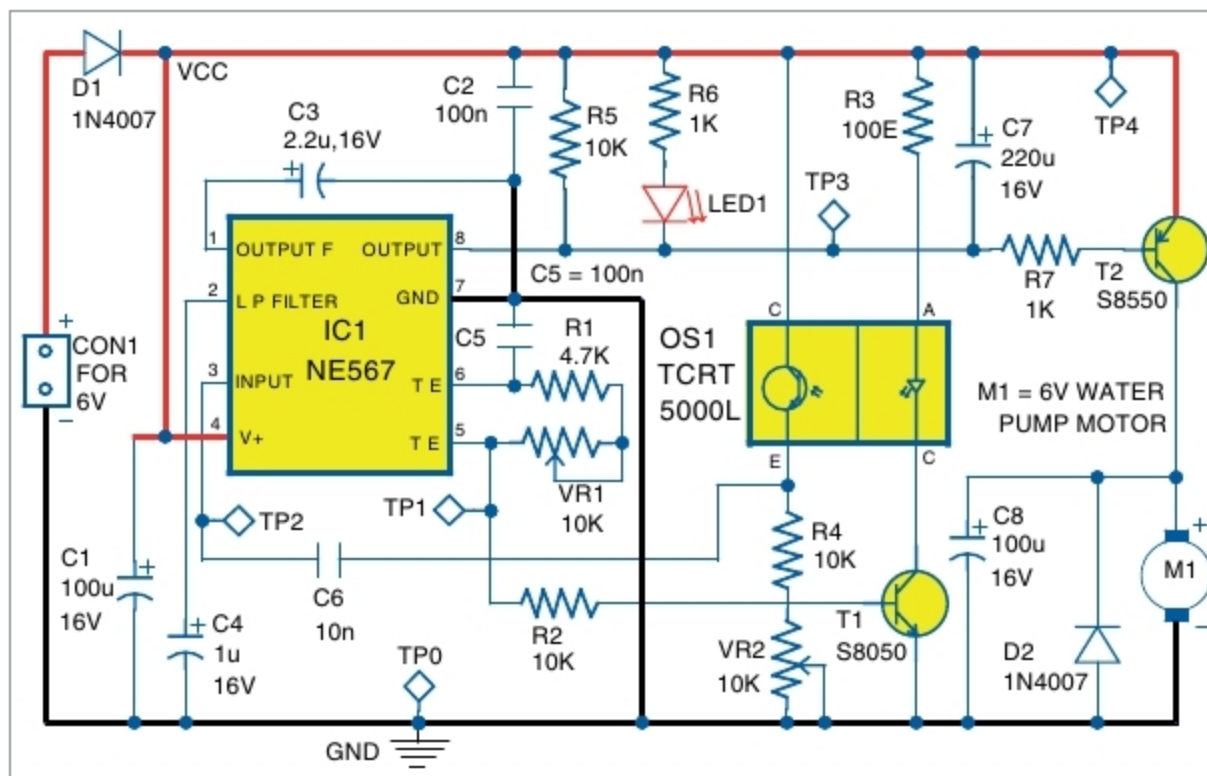


Fig. 2: Circuit diagram of the automatic water dispenser